# Practice 1: Machine Learning for the detection of speech pathologies

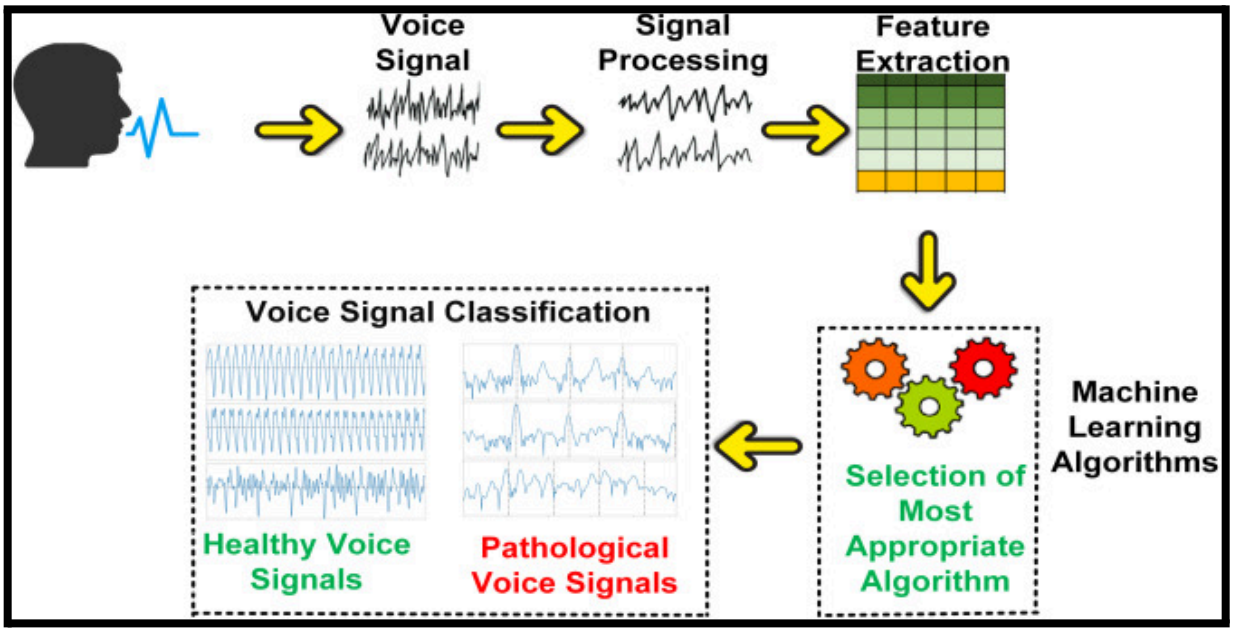



Image from: M. U. Rehman et al., "Voice disorder detection using machine learning algorithms: An application in speech and language pathology

Carla Berard Gonzalez

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# 1) Introduction to the problem

Voice disorders can seriously affect a person's ability to communicate and their daily life. Usually, diagnosing these conditions involves clinical procedures like laryngoscopy or expert auditory evaluation. Although these methods are effective, they require specialized equipment and trained professionals, which makes the process longer, more expensive, and not always easily accessible. For this reason, there is growing interest in digital tools that can help detect voice problems in a faster, more objective, and non-invasive way.

In this project, I explore the use of audio processing and machine learning to automatically classify voices as healthy or pathological. The dataset includes recordings from both healthy speakers and patients with different voice disorders. From each recording, I extract relevant acoustic features, such as MFCCs, Zero-Crossing Rate (ZCR), and RMS energy, to better represent the characteristics of the voice signal.

Using these features, I train and test multiple machine-learning models to evaluate how well they can distinguish between healthy and pathological voices. I then compare their performance to identify which model works best. The goal is not only to achieve high accuracy, but also to make sure the model performs well in realistic conditions. This includes handling data variability (because real patients' voices can differ a lot in age, recording quality, accent, and speaking style) and ensuring that the model can generalize well, meaning it should work properly not only on the training data, but also on new, unseen recordings.

It is important to mention that I am approaching this as a student, not as a clinical or machine-learning professional. My objective is to apply the methods learned in class and do my best to achieve the most accurate and realistic results possible within this academic context. Ultimately, the aim is to explore whether machine-learning could support early detection of voice disorders in a practical, reliable, and accessible way.

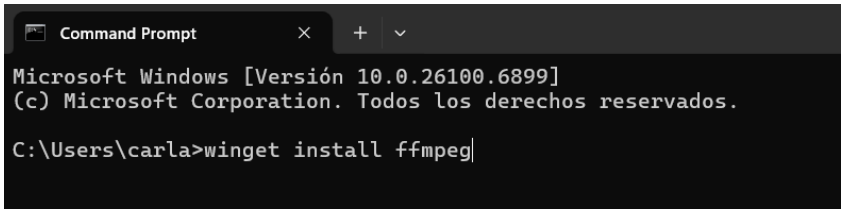
## 2) Development of the project

# PART I

After doing some research, I first decided to work with the EGG (Electroglottograph) signal. The EGG directly measures the contact and separation of the vocal folds during phonation by detecting tiny changes in electrical impedance across the larynx. Since voice disorders often affect how the vocal folds vibrate and close, this signal provides a direct and physiologically meaningful representation of vocal-fold behavior, which makes it particularly suitable for voice pathology analysis. **[2]**

The dataset also includes NSP (Neck Surface) recordings, which capture the vibrations on the neck surface above the larynx. Although NSP signals contain valuable acoustic information related to phonation, they are more susceptible to external influences compared to EGG. Neck tissues can filter the vibration, microphone placement can vary across recordings, and ambient noise or slight movements may interfere with the signal quality **[1]**. Therefore, while NSP reflects the acoustic outcome of vocal-fold vibration, it is considered less clean and less directly tied to the physiological voice mechanism. Because of this, I initially prioritized the EGG signal as a more reliable source for detecting vocal-fold abnormalities.

Once the EGG files were selected, I balanced the classes to prevent bias toward any specific pathology group. Since the original recordings were provided in .egg format, I converted them to a standard .wav format in order to use librosa's library. I installed the FFmpeg executable (to do the conversion) and processed the files through the command prompt to perform the batch conversion. This allowed me to work with the audio using standard Python libraries and proceed with the feature extraction and machine-learning pipeline.



## Exploratory analysis and feature selection

Based on previous studies and the characteristics of the EGG signal, I initially selected a set of time- and frequency-domain features to capture key aspects of vocal behavior.

Time-domain features

- **RMS Energy:** Represents the overall signal strength and vocal-fold closure. Higher RMS values typically indicate healthy phonation, while lower values are associated with weak or breathy voice quality **[6]**.

- **Zero-Crossing Rate (ZCR):** Measures how often the signal crosses zero amplitude. Higher ZCR usually reflects instability or noise in the voice signal, a common indicator of dysphonia **[7]**.

Frequency-domain features

- **Spectral Centroid:** Describes the "brightness" of the signal. Higher values often appear in tense or strained phonation **[8]**.

- **Spectral Flatness:** Indicates how noise-like the signal is. Healthy voices tend to be more harmonic (lower flatness), while pathological voices show flatter and more irregular spectra **[9]**.

- **MFCCs:** Capture the overall spectral shape of the voice and are widely used as reliable features for distinguishing healthy from pathological phonation **[10]**.

To avoid doing manual or subjective feature selection, I decided to use a data-driven filter method with SelectKBest from scikit-learn. Filter methods are simple and objective because they look at how strongly each feature is related to the target variable, without depending on any specific model. In my case, the ANOVA F-test measured how strongly each feature distinguishes between the classes before training the models. SelectKBest tests each feature individually, assigns it an F-value score, ranks all the features, and then keeps only the top-K most informative ones.

This helps remove irrelevant or noisy features that do not contribute much to classification, reducing overfitting and improving generalization **[17]**.

I used different K values for each model depending on how they handle the amount of input information

- For Logistic Regression, I set k = 30 because linear models tend to work better with a moderate number of the strongest predictors to avoid instability and overfitting **[18]**.
- For the Decision Tree, I used k = 10, since trees can easily overfit if they see too many variables, and focusing on fewer but more discriminative ones helps them generalize better **[19]**.
- For Random Forest and Gradient Boosting, I kept all features (k = None), because ensemble models are more robust, they automatically ignore unimportant variables through random sampling and iterative training **[20]**.

I chose this filter approach instead of wrapper methods because wrappers train the model many times with different feature combinations, which is computationally expensive and can lead to overfitting.

I also used a patient-wise train–test split (≈75% train, 25% test) so that all recordings from a given patient appear only in one set. This ensures that the final test results come from completely new patients that the model has never seen before.

## Explanation of the machine learning models I have trained

In this project I trained four models because each one represents a different way of learning from the voice features, and comparing them helps me understand which type works best for this classification task:

Logistic Regression was added as a simple baseline that is commonly used in voice-pathology studies [11][12]. It does not try to learn complex shapes, so it shows how a basic linear classifier performs with these features.

I also added a Decision Tree because it is the simplest rule based model. It tests whether the classes can be separated using simple if-else splits on features like RMS, ZCR, or MFCCs. If the tree performs poorly, it already suggests the problem requires more sophisticated models.

Finally, I used Random Forest and Gradient Boosting because they are stronger ensemble methods. Random Forest averages many trees and is robust for biomedical signals [15][16], while Gradient Boosting learns from errors step by step and can capture more subtle voice differences [20].

## Discussion and validation of the results

Accuracy

Accuracy tells us how many samples the model classified correctly out of the total. In this part, the best-performing model was Gradient Boosting, with an accuracy of 0.433, meaning it correctly classified about 43% of the EGG recordings. Logistic Regression (0.428) and Random Forest (0.427) performed almost identically, while the Decision Tree had the lowest performance (0.320).

Balanced Accuracy

Balanced Accuracy checks how well the model performs across all diseases equally, instead of just doing well on the easiest one. The Gradient Boosting model again was strongest (0.432), followed closely by Logistic Regression and Random Forest (~0.427). The Decision Tree performed worst (~0.320).

A key observation is that accuracy and balanced accuracy are almost identical for all models. This means the models did not focus on one disease and ignore others, in other words, performance was balanced across disorders, which is crucial in medical diagnosis where every condition matters.

Macro-F1 Score

Macro-F1 combines two ideas:

- Precision: when the model predicts a disease, how often is it right?
- Recall: how many real cases of the disease did it actually detect?

At first, I considered using the regular F1 score, but I decided to use the Macro-F1 instead because even if the classes are balanced in quantity, they are not equally easy to classify. For instance, Laryngitis may have very distinct vocal features, making it easier for the model to recognize, while Hyperfunctional Dysphonia can be much subtler and harder to detect. If I used the standard F1 score, the easier classes could dominate the metric and make the results look better than they really are.

Macro-F1 treats all classes equally, it calculates the F1 score for each disease and then averages them, giving the same importance to both easy and difficult cases. This makes it a fairer and more meaningful metric in medical contexts, where the goal is not only to do well overall but also to detect every pathology as accurately as possible.

The best Macro-F1 was again achieved by Gradient Boosting (0.4286), meaning it managed both precision and recall reasonably well across all classes. Logistic Regression and Random Forest followed closely (~0.42), while the Decision Tree remained the weakest (~0.32).

Confusion Matrices (Raw vs Normalized)

I used raw and normalized confusion matrices to analyze predictions:

- Raw matrix = exact number of samples predicted correctly/incorrectly
- Normalized matrix = shows percentages per class (per-class recall)

This helps us see which disorders are harder or easier to detect.

Findings (taking into account best accuracies, GB, LG and RF):

- Healthy and Laryngitis were classified most accurately
  ~0.54–0.58 recall for Healthy
  ~0.49–0.50 recall for Laryngitis
- Hyperfunctional Dysphonia was the hardest (~0.30 recall), likely because its glottal symptoms are more subtle and variable
- Recurrent Laryngeal Nerve Paralysis sat in the middle (~0.34–0.39 recall)

This means the models had difficulty with certain pathologies, but did not show systematic bias toward a specific class.

Why Gradient Boosting performed best

Gradient Boosting achieved the best performance across all metrics. This suggests that EGG features contain mostly linear but slightly nonlinear patterns. Logistic Regression was almost as good, confirming that much of the class separation is linear. However, Gradient Boosting improved performance by capturing additional subtle nonlinearities.

# PART II

## Exploratory analysis and feature selection (II)

After running the first experiments with my initial feature set (RMS, ZCR, spectral centroid, spectral flatness, and MFCCs), I realized that my selection, although supported by literature, was still subjective. It did not necessarily represent the best combination of features for every machine-learning model. To make the process more objective and data-driven, I expanded the feature-extraction process and allowed the data to guide the selection rather than relying only on prior knowledge.

In addition to the previously used features, I extracted a much broader set of acoustic descriptors to capture more aspects of the voice signal. Specifically, I added:

- **Spectral bandwidth**: reflects the spread of frequencies around the spectral centroid.
- **Spectral roll-off**: indicates the frequency below which a fixed percentage of the total spectral energy lies (useful for differentiating harmonic vs. noisy signals).
- **Chroma features**: represent the energy distribution across the 12 pitch classes, complementing MFCCs by focusing on harmonic content.
- **Spectral contrast**: measures the difference between peaks and valleys in the spectrum, helping to distinguish harmonic structure from noise-dominated regions.
- **Jitter estimates and f0 variability**: capturing pitch instability, which is characteristic of many voice disorders.

Apart from using SelectKBest(f_classif) to rank the features, I made the process more reliable by combining it with GroupKFold cross-validation and GridSearchCV.

The idea was to let the data decide how many features each model should use instead of fixing k manually.

Then, inside the training set, I applied GroupKFold.

This method creates several mini train–validation splits (like smaller internal trainings), but always keeps all recordings from the same patient together in one fold.

So during these internal rounds, the model is trained on some training patients and validated on other training patients, never mixing data from the same person. This is done multiple times, which is what makes it “cross-validation.”

Within this process, GridSearchCV automatically tested different numbers of selected features (k = 10, 30, or all 54) for each model and chose the value of k that achieved the best average validation accuracy.

After finding the best k for each classifier (Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting), I retrained that model on the entire training set and finally evaluated it once on the completely new test patients.

In summary, combining SelectKBest, GroupKFold, and GridSearchCV made the whole feature selection process much more robust and objective.

The patient-wise split guaranteed a fair final evaluation on unseen patients, while the GroupKFold inside the training set allowed safe internal cross-validation without any data leakage between patients.

## Discussion and validation of the results (II)

After applying a GroupKFold cross-validation strategy together with SelectKBest for data-driven feature selection, model performance showed small but consistent improvements in some algorithms, while others remained stable or slightly decreased.

In the earlier setup, where only handcrafted features such as RMS, ZCR, spectral centroid, spectral flatness, and MFCCs were used, Gradient Boosting achieved the highest accuracy (0.433), followed by Logistic Regression (0.428) and Random Forest (0.427). After expanding the feature set and introducing automatic feature selection, Random Forest became the best model, improving slightly from 0.427 to 0.442, while Gradient Boosting performed slightly worse (0.433 → 0.424). Logistic Regression also decreased (0.428 → 0.413), suggesting that it benefits less from a larger feature space. Decision Tree remained the weakest, with similar results (~0.320 → ~0.335), confirming its tendency to overfit and its limited ability to generalize.

Balanced accuracy and macro-F1 values followed the same pattern as accuracy, showing consistent behavior across pathology classes and no clear bias toward any specific condition.

The confusion matrices confirm these observations. Healthy and Laryngitis were the easiest to identify, with recall values around 0.54–0.56 and 0.50–0.52, respectively. Hyperfunctional Dysphonia remained the most difficult class (~0.30–0.31 recall), likely due to its subtle and variable voice patterns, while Recurrent Laryngeal Nerve Paralysis achieved intermediate recall (~0.37–0.39).

These trends were very similar to those observed in the previous experiment, indicating that while the expanded feature set and GroupKFold/SelectKBest pipeline slightly improved overall robustness, the relative difficulty of each pathology class remained unchanged.

# PART III

## Exploratory analysis and feature selection (III)

After finishing the experiments with the EGG recordings, I decided to try the same approach but this time only with the NSP files. Even though both signals relate to vocal-fold activity, they are fundamentally different:

- **EGG** measures how the vocal folds open and close by detecting the contact between them.
- **NSP** on the other hand, records the sound vibrations on the neck, so it captures things like resonance, breathiness and harmonic structure.

This means NSP does not measure fold contact as precisely as EGG, but it offers richer acoustic and articulatory information that could reveal additional clues about pathology. Because these modalities are complementary, focusing next on NSP allowed me to study whether this additional acoustic richness could improve classification performance.

For the NSP experiment I used the same expanded acoustic feature set that I used in Part II, and inside the training set I again used GroupKFold to tune the SelectKBest parameter.

In the first EGG experiments, I treated each recording (each vowel) as a separate sample. So if one patient had several recordings, that patient appeared multiple times in the dataset. Even though I used GroupKFold, the model was still learning per recording, not per patient. For the NSP analysis, I changed this. I wanted each patient to appear only once, so I combined all their recordings and averaged all their features using pandas groupby.

This means that instead of having one row per recording, the dataset now has one single row per patient, containing the average of their acoustic characteristics.This makes the data more stable and reduces noise caused by differences in vowels or articulation. It also reflects real clinical practice better, since diagnoses are made per patient rather than per individual recording.

## Discussion and validation of the results (III)

Model performance with NSP features was broadly similar to the EGG results, with some models slightly better and others worse. The best NSP model was Logistic Regression (0.440 accuracy), which is higher than its EGG score (0.413 → 0.440). Random Forest performed similarly but slightly decreased (0.442 → 0.426). Gradient Boosting also showed a small drop (0.424 → 0.404), while the Decision Tree slightly improved (~0.335 → 0.355). In all cases, balanced accuracy and macro-F1 closely followed accuracy, confirming that the models treated all pathology classes fairly and did not favor any specific condition.

The confusion matrices show patterns very similar to the EGG experiments. Laryngitis remained the easiest class to recognize, with recall values around 0.53–0.61, followed by Recurrent Laryngeal Nerve Paralysis with intermediate recall (0.34–0.43). Healthy voices were sometimes misclassified as mildly pathological (0.31–0.43 recall) and Hyperfunctional Dysphonia remained the most difficult class to identify (~0.23–0.40 recall), consistent with its subtle and variable acoustic presentation. These consistent patterns across both EGG and NSP indicate that classification difficulty depends more on the nature of the pathology than on the recording modality.

The most relevant NSP features were spectral contrast, chroma, and several MFCCs, all related to resonance and harmonic structure. This aligns with the physical difference between both signals: NSP captures vocal-tract resonances and radiated sound, while EGG measures vocal-fold contact directly. Each modality thus provides complementary physiological information.

Overall, the NSP results confirm that neck-acoustic recordings can detect voice disorders about as effectively as EGG.

# PART IV

## Exploratory analysis and feature selection (IV)

After finishing the EGG and NSP experiments separately, I decided to try combining both signals to see if the models could perform better. Since EGG and NSP capture different but related information (EGG measures vocal-fold contact and NSP captures acoustic resonance from the neck), using them together might give the model a more complete picture of the voice and help it detect pathologies more accurately.

To combine them, I tested two simple strategies:

1) Feature-level fusion (early fusion)

Here I joined both feature sets before training. Since I already had one feature vector per patient for EGG and one for NSP, I just matched patients by their ID and merged both feature tables side-by-side.

So each patient has one long feature vector that includes EGG + NSP features together. Then I trained the models on this combined dataset.

This lets the model learn everything at once, both the glottal information from EGG and the acoustic information from NSP.

2) Score-level fusion (late fusion)

In this approach, I trained one model using only EGG features and another one using only NSP features.

Then, instead of merging the features, I merged the predictions by averaging the probability outputs from both models and picking the final class based on that combined score.

## Discussion and validation of the results (IV)

The combined approach gave the best results of the whole project. When I fused the EGG and NSP features, the Random Forest reached 0.504 accuracy (NSP 0.426 → Fusion 0.504; EGG 0.442 → Fusion 0.504). Gradient Boosting also improved to 0.482 (EGG 0.424 → 0.482; NSP 0.404 → 0.482).

Logistic Regression stayed close to its previous levels at 0.433 (EGG 0.413 → 0.433; NSP 0.440 → 0.433).

The Decision Tree remained the weakest at 0.383 (EGG ~0.335 → 0.383; NSP 0.355 → 0.383).

As before, balanced accuracy and macro-F1 closely tracked accuracy for every model, indicating that fusion improved performance without favoring a single class.

Moreover, the confusion matrices support this. As in the EGG or NSP separate runs, Laryngitis and Recurrent Laryngeal Nerve Paralysis were the easiest classes, and their recalls increased with the fusion (the model can rely on both glottal-contact cues from EGG and resonance/harmonic cues from NSP). Healthy was classified more consistently and Hyperfunctional Dysphonia remained the hardest but still benefited slightly from fusion. Overall class behavior stayed the same, but predictions became more stable and accurate.

Feature importance showed contributions from both modalities, confirming they are complementary rather than redundant. Combining them yields a richer representation of each patient's voice.

# PART V

For the live prediction, I only used the file Speech_pathologies_dataset_NSP_features.csv, because this dataset contains NSP (neck-surface or acoustic) features that can be obtained directly from a regular microphone.

The EGG (electroglottographic) features, on the other hand, come from a special neck sensor that measures vocal fold contact. Both EGG and NSP are highly useful and complementary for research and diagnosis, but for the live prediction test, NSP is the most suitable option. This is because the NSP modality captures acoustic information, the sound vibrations that naturally radiate through the skin and air, so it can be measured non-invasively and in real time using any standard microphone.

This makes it ideal for building a portable, accessible system that can predict voice conditions without requiring specialized sensors.

During the live test, the script recorded the user's voice through the microphone in mono format at 44.1 kHz, which is standard quality for speech. Each session included three short recordings of 5 seconds each, where the user was asked to sustain a steady vowel sound (such as "aaaaa"). The program automatically recorded each take and then averaged the extracted features across all three to make the result more stable and consistent, minimizing random noise or short-term variations in the voice.

Before extracting the features, the audio went through an automatic cleaning process:

- Pre-emphasis: slightly boosts high frequencies to make the spectrum more balanced.
- Soft band-pass filtering (70–4000 Hz): keeps the main speech frequencies and removes unwanted low or high noise.
- Trimming and normalization: removes silence at the beginning and end, and normalizes loudness.

After cleaning, the script extracted different types of acoustic features:

- Time-domain: RMS energy and Zero-Crossing Rate (ZCR).
- Cepstral features: the first 13 MFCCs (Mel-Frequency Cepstral Coefficients), describing how sound energy is distributed across frequencies.
- Harmonic and resonance features: Chroma and Spectral Contrast, which capture tone and harmonic richness.
- Frequency-domain: spectral centroid, bandwidth, flatness, and rolloff.

All extracted features were then arranged with the same column names as in the NSP dataset so that the trained model (trained_model_NSP.pkl) could process them directly. The program automatically aligned the feature order, made a prediction, and displayed both the predicted voice condition and the confidence level.

If the model predicted a pathology with a confidence below 60%, it displayed a warning suggesting that the user repeat the test in a quieter environment with a more stable vowel sound.

# 3)Conclusions

At the beginning, this project was quite challenging because I had to apply concepts that were still quite new, like feature selection, model evaluation, and data preprocessing. We had only recently learned these topics in class, so putting them into practice in a real project took a lot of research, trial and error, and patience to make everything work properly.

Little by little, it became clearer how the features, models, and evaluation metrics work together to detect voice pathologies. I realized that the goal wasn't only to get the highest accuracy, but to make sure the models were fair and reliable across all classes.

Even though the results weren't perfect, the experiments showed that both EGG and NSP contain useful information, and that combining them improves performance. I also learned that the NSP signal is the most practical one for real-life use, since it can be recorded with a normal microphone.

Overall, I'm happy with how much I learned, not only about coding and machine learning, but also about how to think critically, be persistent, and connect theory with practice. It wasn't easy, but it was a really good experience to see how everything we learned in class can actually be applied to a real biomedical problem.

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